	$\mathcal{K}_{0^+}^{\#1}$	$\mathcal{K}_{0^-}^{\#1\,\alpha\beta\chi}$
$\mathcal{K}_{0^+}^{\#1\,\dagger}$	$\frac{1}{2}(-Y_1 k^2 - 3\,\overset{(2)}{\kappa}_3)$	0
$\mathcal{K}_{0^-}^{\#1\,\dagger\alpha\beta\chi}$	0	0

	$\mathcal{K}_{1^+}^{\#1\,\alpha\beta}$	$\mathcal{K}_{1^+}^{\#2\,\alpha\beta}$	$\mathcal{K}_{1^-}^{\#1\,\alpha}$	$\mathcal{K}_{1^-}^{\#2\,\alpha}$
$\mathcal{K}_{1^+}^{\#1\,\dagger\alpha\beta}$	$-\frac{Y_2 k^2}{2}$	$\frac{k^2\,\overset{(4)}{\kappa}_4}{2\sqrt{2}}$	0	0
$\mathcal{K}_{1^+}^{\#2\,\dagger\alpha\beta}$	$\frac{k^2\,\overset{(4)}{\kappa}_4}{2\sqrt{2}}$	$\frac{Y_3 k^2}{2}$	0	0
$\mathcal{K}_{1^-}^{\#1\,\dagger\alpha}$	0	0	$-k^2\,\overset{(4)}{\kappa}_2 - \overset{(2)}{\kappa}_3$	$\frac{k^2\,\overset{(4)}{\kappa}_2 + \overset{(2)}{\kappa}_3}{\sqrt{2}}$
$\mathcal{K}_{1^-}^{\#2\,\dagger\alpha}$	0	0	$\frac{k^2\,\overset{(4)}{\kappa}_2 + \overset{(2)}{\kappa}_3}{\sqrt{2}}$	$\frac{1}{2}(-k^2\,\overset{(4)}{\kappa}_2 - \overset{(2)}{\kappa}_3)$

	$\mathcal{K}_{2^+}^{\#1\,\alpha\beta}$	$\mathcal{K}_{2^-}^{\#1\,\alpha\beta\chi}$
$\mathcal{K}_{2^+}^{\#1\,\dagger\alpha\beta}$	$-\frac{Y_4 k^2}{2}$	0
$\mathcal{K}_{2^-}^{\#1\,\dagger\alpha\beta\chi}$	0	0

	$\mathcal{J}_{0^+}^{\#1}$	$\mathcal{J}_{0^-}^{\#1\,\alpha\beta\chi}$
$\mathcal{J}_{0^+}^{\#1\,\dagger}$	$\frac{1}{\text{Det}(0^+)}$	0
$\mathcal{J}_{0^-}^{\#1\,\dagger\alpha\beta\chi}$	0	0

	$\mathcal{J}_{1^+}^{\#1\,\alpha\beta}$	$\mathcal{J}_{1^+}^{\#2\,\alpha\beta}$	$\mathcal{J}_{1^-}^{\#1\,\alpha}$	$\mathcal{J}_{1^-}^{\#2\,\alpha}$
$\mathcal{J}_{1^+}^{\#1\,\dagger\alpha\beta}$	$\frac{Y_3 k^2}{2\,\text{Det}(1^+)}$	$-\frac{k^2\,\overset{(4)}{\kappa}_4}{2\sqrt{2}\,\text{Det}(1^+)}$	0	0
$\mathcal{J}_{1^+}^{\#2\,\dagger\alpha\beta}$	$-\frac{k^2\,\overset{(4)}{\kappa}_4}{2\sqrt{2}\,\text{Det}(1^+)}$	$-\frac{Y_2 k^2}{2\,\text{Det}(1^+)}$	0	0
$\mathcal{J}_{1^-}^{\#1\,\dagger\alpha}$	0	0	$\frac{2}{3\,\text{Det}(1^-)}$	$-\frac{\sqrt{2}}{3\,\text{Det}(1^-)}$
$\mathcal{J}_{1^-}^{\#2\,\dagger\alpha}$	0	0	$-\frac{\sqrt{2}}{3\,\text{Det}(1^-)}$	$\frac{1}{3\,\text{Det}(1^-)}$

	$\mathcal{J}_{2^+}^{\#1\,\alpha\beta}$	$\mathcal{J}_{2^-}^{\#1\,\alpha\beta\chi}$
$\mathcal{J}_{2^+}^{\#1\,\dagger\alpha\beta}$	$\frac{1}{\text{Det}(2^+)}$	0
$\mathcal{J}_{2^-}^{\#1\,\dagger\alpha\beta\chi}$	0	0

Abbreviations used in matrices

$$Y_1 == 3\,\overset{(4)}{\kappa}_1 + 3\,\overset{(4)}{\kappa}_2 + \overset{(4)}{\kappa}_3 + \overset{(4)}{\kappa}_6 \;\&\&\; Y_2 == \overset{(4)}{\kappa}_3 - \overset{(4)}{\kappa}_6 \;\&\&\; Y_3 == \overset{(4)}{\kappa}_3 + \overset{(4)}{\kappa}_4 \;\&\&$$

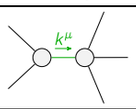
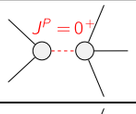
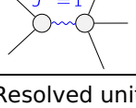
$$Y_4 == \overset{(4)}{\kappa}_3 + \overset{(4)}{\kappa}_6 \;\&\&\; \text{Det}(0^+) == \frac{1}{2}(-k^2(3\,\overset{(4)}{\kappa}_1 + 3\,\overset{(4)}{\kappa}_2 + \overset{(4)}{\kappa}_3 + \overset{(4)}{\kappa}_6) - 3\,\overset{(2)}{\kappa}_3) \;\&\&$$

$$\text{Det}(1^+) == -\frac{1}{8}k^4(2\,\overset{(4)}{\kappa}_3{}^2 + 2\,\overset{(4)}{\kappa}_3\,\overset{(4)}{\kappa}_4 + \overset{(4)}{\kappa}_4{}^2 - 2(\overset{(4)}{\kappa}_3 + \overset{(4)}{\kappa}_4)\,\overset{(4)}{\kappa}_6) \;\&\&\; \text{Det}(2^+) == -\frac{1}{2}k^2(\overset{(4)}{\kappa}_3 + \overset{(4)}{\kappa}_6)$$

Lagrangian

$$\overset{(2)}{\kappa}_3\,\mathcal{K}^{\alpha}{}_{\alpha}{}^{\beta}\,\mathcal{K}^{\chi}{}_{\beta\chi} + \overset{(4)}{\kappa}_1\,\partial_\beta\mathcal{K}^{\delta}{}_{\chi\delta}\,\partial^\chi\mathcal{K}^{\alpha}{}_{\alpha}{}^{\beta} + \overset{(4)}{\kappa}_2\,\partial_\chi\mathcal{K}^{\delta}{}_{\beta\delta}\,\partial^\chi\mathcal{K}^{\alpha}{}_{\alpha}{}^{\beta} + \overset{(4)}{\kappa}_3\,\partial_\beta\mathcal{K}^{\alpha\beta\chi}\,\partial_\delta\mathcal{K}_{\alpha\chi}{}^{\delta} +$$
$$\overset{(4)}{\kappa}_4\,\partial_\alpha\mathcal{K}^{\alpha\beta\chi}\,\partial_\delta\mathcal{K}_{\beta\chi}{}^{\delta} + \overset{(4)}{\kappa}_6\,\partial_\beta\mathcal{K}^{\alpha\beta\chi}\,\partial_\delta\mathcal{K}_{\chi\alpha}{}^{\delta} + \frac{1}{2}\,\overset{(4)}{\kappa}_3\,\partial_\alpha\mathcal{K}^{\alpha\beta\chi}\,\partial_\delta\mathcal{K}^{\delta}{}_{\beta\chi} + \frac{1}{2}\,\overset{(4)}{\kappa}_4\,\partial_\alpha\mathcal{K}^{\alpha\beta\chi}\,\partial_\delta\mathcal{K}^{\delta}{}_{\beta\chi}$$

Added source term(s):	$\mathcal{K}^{\alpha\beta\chi}\;\mathcal{J}_{\alpha\beta\chi}$	
Source constraint(s)	# constraint(s)	Covariant form
$\mathcal{J}_{0^-}^{\#1\,\alpha\beta\chi} == 0$	1	$\partial_\delta\partial^\alpha\mathcal{J}^{\beta\chi\delta} + \partial_\delta\partial^\alpha\mathcal{J}^{\delta\beta\chi} + \partial_\delta\partial^\beta\mathcal{J}^{\chi\alpha\delta} + \partial_\delta\partial^\chi\mathcal{J}^{\alpha\beta\delta} + \partial_\delta\partial^\chi\mathcal{J}^{\delta\alpha\beta} + \partial_\delta\partial^\delta\mathcal{J}^{\beta\alpha\chi} ==$ $\partial_\delta\partial^\alpha\mathcal{J}^{\chi\beta\delta} + \partial_\delta\partial^\beta\mathcal{J}^{\alpha\chi\delta} + \partial_\delta\partial^\beta\mathcal{J}^{\delta\alpha\chi} + \partial_\delta\partial^\chi\mathcal{J}^{\beta\alpha\delta} + \partial_\delta\partial^\delta\mathcal{J}^{\alpha\beta\chi} + \partial_\delta\partial^\delta\mathcal{J}^{\chi\alpha\beta}$
$\mathcal{J}_{1^-}^{\#1\,\alpha} + 2\,\mathcal{J}_{1^-}^{\#2\,\alpha} == 0$	3	$\partial_\chi\partial^\alpha\mathcal{J}^{\beta}{}_{\beta}{}^{\chi} + \partial_\chi\partial^\chi\mathcal{J}^{\beta\alpha}{}_{\beta} == 3\,\partial_\chi\partial_\beta\mathcal{J}^{\beta\alpha\chi}$
$\mathcal{J}_{2^-}^{\#1\,\alpha\beta\chi} == 0$	5	$3\,\partial_\epsilon\partial_\delta\partial^\chi\partial^\alpha\mathcal{J}^{\delta\beta\epsilon} + 3\,\partial_\epsilon\partial^\epsilon\partial^\chi\partial^\alpha\mathcal{J}^{\delta\beta}{}_{\delta} + 2\,\partial_\epsilon\partial^\epsilon\partial_\delta\partial^\beta\mathcal{J}^{\chi\alpha\delta} + 4\,\partial_\epsilon\partial^\epsilon\partial_\delta\partial^\beta\mathcal{J}^{\chi\alpha\delta} + 2\,\partial_\epsilon\partial^\epsilon\partial_\delta\partial^\beta\mathcal{J}^{\delta\alpha\chi} + 2\,\partial_\epsilon\partial^\epsilon\partial_\delta\partial^\chi\mathcal{J}^{\beta\alpha\delta} +$ $4\,\partial_\epsilon\partial^\epsilon\partial_\delta\partial^\chi\mathcal{J}^{\delta\alpha\beta} + 2\,\partial_\epsilon\partial^\epsilon\partial_\delta\partial^\delta\mathcal{J}^{\alpha\beta\chi} + 3\,\eta^{\beta\chi}\,\partial_\phi\partial^\phi\partial_\epsilon\partial^\alpha\mathcal{J}^{\delta}{}_{\delta}{}^{\epsilon} + 3\,\eta^{\alpha\chi}\,\partial_\phi\partial^\phi\partial_\epsilon\partial_\delta\mathcal{J}^{\delta\beta\epsilon} + 3\,\eta^{\beta\chi}\,\partial_\phi\partial^\phi\partial_\epsilon\partial^\epsilon\mathcal{J}^{\delta\alpha}{}_{\delta} ==$ $3\,\partial_\epsilon\partial_\delta\partial^\chi\partial^\beta\mathcal{J}^{\delta\alpha\epsilon} + 3\,\partial_\epsilon\partial^\epsilon\partial^\chi\partial^\beta\mathcal{J}^{\delta\alpha}{}_{\delta} + 2\,\partial_\epsilon\partial^\epsilon\partial_\delta\partial^\alpha\mathcal{J}^{\beta\chi\delta} + 4\,\partial_\epsilon\partial^\epsilon\partial_\delta\partial^\alpha\mathcal{J}^{\chi\beta\delta} + 2\,\partial_\epsilon\partial^\epsilon\partial_\delta\partial^\alpha\mathcal{J}^{\delta\beta\chi} + 2\,\partial_\epsilon\partial^\epsilon\partial_\delta\partial^\chi\mathcal{J}^{\alpha\beta\delta} +$ $2\,\partial_\epsilon\partial^\epsilon\partial_\delta\partial^\delta\mathcal{J}^{\beta\alpha\chi} + 4\,\partial_\epsilon\partial^\epsilon\partial_\delta\partial^\delta\mathcal{J}^{\chi\alpha\beta} + 3\,\eta^{\alpha\chi}\,\partial_\phi\partial^\phi\partial_\epsilon\partial^\beta\mathcal{J}^{\delta}{}_{\delta}{}^{\epsilon} + 3\,\eta^{\beta\chi}\,\partial_\phi\partial^\phi\partial_\epsilon\partial_\delta\mathcal{J}^{\delta\alpha\epsilon} + 3\,\eta^{\alpha\chi}\,\partial_\phi\partial^\phi\partial_\epsilon\partial^\epsilon\mathcal{J}^{\delta\beta}{}_{\delta}$
Total # constraint(s):	9	

Resolved pole(s)	# polarization(s)	Square mass	Residue
	2	0	$-\frac{\overset{(4)}{\kappa}_3{}^2 + \overset{(4)}{\kappa}_3\,\overset{(4)}{\kappa}_4 + \overset{(4)}{\kappa}_4{}^2 - 3\,\overset{(4)}{\kappa}_4\,\overset{(4)}{\kappa}_6 + 3\,\overset{(4)}{\kappa}_6{}^2}{(2\,\overset{(4)}{\kappa}_3{}^2 + \overset{(4)}{\kappa}_4(\overset{(4)}{\kappa}_4 - 2\,\overset{(4)}{\kappa}_6) + 2\,\overset{(4)}{\kappa}_3(\overset{(4)}{\kappa}_4 - \overset{(4)}{\kappa}_6))(\overset{(4)}{\kappa}_3 + \overset{(4)}{\kappa}_6)}$
	1	$-\frac{3\,\overset{(2)}{\kappa}_3}{3\,\overset{(4)}{\kappa}_1 + 3\,\overset{(4)}{\kappa}_2 + \overset{(4)}{\kappa}_3 + \overset{(4)}{\kappa}_6}$	$-\frac{2}{3\,\overset{(4)}{\kappa}_1 + 3\,\overset{(4)}{\kappa}_2 + \overset{(4)}{\kappa}_3 + \overset{(4)}{\kappa}_6}$
	3	$-\frac{\overset{(2)}{\kappa}_3}{\overset{(4)}{\kappa}_2}$	$\frac{2}{3\,\overset{(4)}{\kappa}_2}$
Resolved unitarity condition(s):	(Demonstrably impossible)		